Data 227 - Autumn 2026 HW 2

Data Visualization and Communication - Trimble

Due: Thursday October 15, 2023 11:59pm

Homework 2, Hurricane tracks

1. The `storms.csv` file contains a subset of the NOAA Atlantic hurricane database best track data. The data includes the positions and attributes of named North Atlantic tropical storms from 1975-2020, measured every six hours during the lifetime of a storm.

   (a) Check the storage type of each variable and state whether it matches your expectations from reading the dataset documentation (i.e., are categorical variables saved as strings? Make sure dates are handled in a format that makes sense (not nominal!)

   (b) Choose a handful of storms according to a theme (perhaps storms in a given season, perhaps storms that leveled a particular location in different years) and perpare a smaller datastructure with just these storms. (Tropical storms live for a few weeks. Don't treat storms with the same name in different years together.)

   (c) Create a scatterplot comparing your storm's maximum sustained windspeed (in knots) to its air pressure at the storm's center (in millibars) for all six-hour measurements. Put the Saffir-Simpson storm catetory into a different visual channel. (note, hurricane/tropical storm status and wind speed, though they may be spread out into multiple columns, measure the same thing.)

   (d) Create a scatterplot (or a map projection!) that shows the track of your collection of tropical storms. **In addition, encode information from tropicalstorm_force_diameter and hurricane_force_diameter** –and if you (very reasonably) want to encode diameter on a **map**, make sure the encoding of storm diameter matches the map scale in at least one dimension.

NHANES

2. The U.S. National Center for Health Statistics has conducted a long-running survey of nutrition and health that recruits about 5000 people a year to be interviewed and subjected to measurements and medical laboratory tests.

   https://wwwn.cdc.gov/nchs/nhanes/default.aspxhttps://wwwn.cdc.gov/nchs/nhanes/default.aspx

   Get data for one or more survey epochs, articulate a question, and generate two visualizations with captions that address your question.

   Reasonable questions within the scope of the survey might include Are older people heavier in aggregate? Do smokers have different incidence of high blood pressure? Are vegetarians shorter than omnivores? Are there gender differences in the diet variables surveyed?

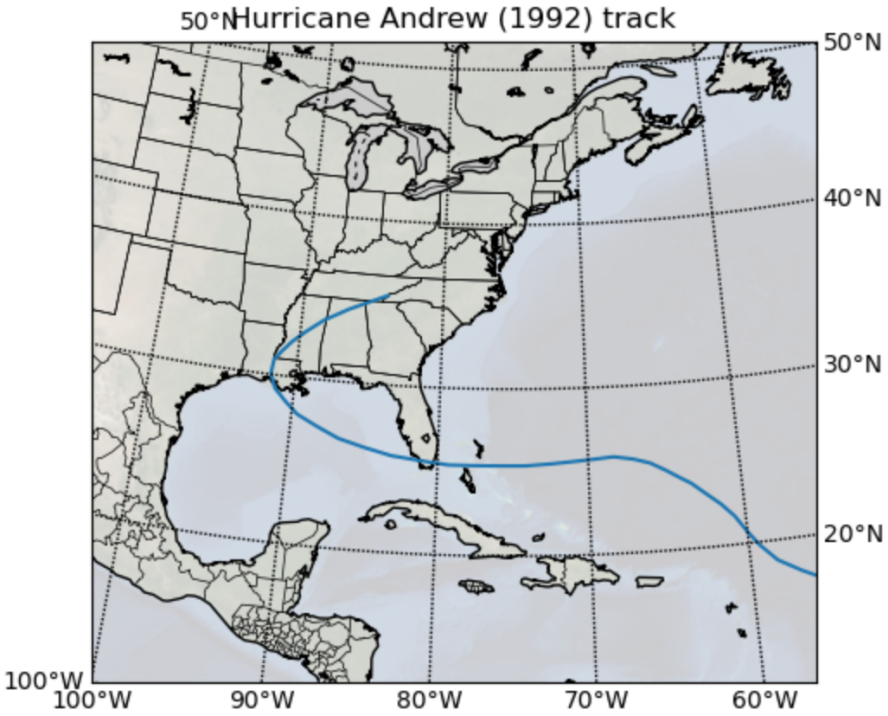


Figure 1: Sample code to produce this map is provided